



Track down RFI with this compact, resistively terminated loop antenna.

Portable Flag Antenna for MF/HF Radio Direction Finding

Don Kirk, WD8DSB

I spend a lot of time tracking down RFI (radio frequency interference) in the MF (medium-frequency) and HF (high-frequency) bands. I often use bi-directional tuned loops, but the portable flag (see the lead image) is unidirectional and broad band, which is a great advantage. For more information, see my YouTube video, at <https://youtu.be/yIWY2KwoZ8w>.

Resistively terminated flags produce a unidirectional cardioid pattern (see Figure 1). Very small flags used on MF and HF have very large losses, but a preamp makes them very useful for direction finding because of their backside null.

My 2-foot-tall by 4-foot-long portable flag is as large as possible — to maximize gain — that would fit inside my car, but still small enough to maintain a cardioid pattern from 1.8 to 30 MHz. With an appropriate preamp, the flag is useful for direction finding even in the AM broadcast band.

General Details

I selected a $\frac{1}{4}$ W non-inductive 680 Ω termination resistor based on modeling in *4nec2* NEC-based antenna modeler (created by Arie Voors, www.qsl.net/4nec2). The front-to-back ratio is typically 12 to 21 dB, which is more than adequate for direction finding.

Figure 2 shows a diagram of my portable flag antenna. I used a BN73-202 binocular core (Fair-Rite 2873000202) transformer at the antenna feed point to match its high impedance to 50 Ω coax. Wind the matching transformer with a three-turn primary and 12-turn secondary using #30 AWG wire-wrap wire. A 10-foot section of RG-174A/U coax connects the matching transformer to the receiver/preamp. Measured SWR is less than 1.5 to 1 from 0.5 to 30 MHz. More details about SWR and gain are on the www.arrl.org/qst-in-depth web page.

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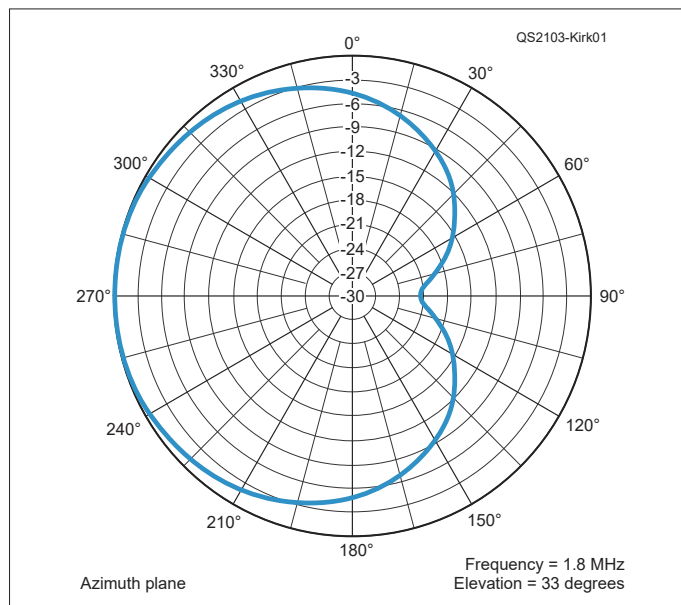


Figure 1 — The cardioid pattern modeled in 4nec2 at the maximum gain elevation angle.

I suppressed HF noise generated by my laptop running SDR software by placing a 31-material clamp-on ferrite core (Fair-Rite part 0431167281) near the end of the coax and winding four passes of the coax through the core (see Figure 3). Placing the ferrite core on the coax closer toward the antenna was less effective. The end of the coax terminates in a shielded phono plug (RadioShack part 274-0339 or Switchcraft part 3502A or equivalent) that connects to the receiver or preamp. I soldered the plug ground lug to the plug body after removing the plating with sandpaper or a file. File down the profile of the solder blob as needed to allow the shielded plug cover to fit.

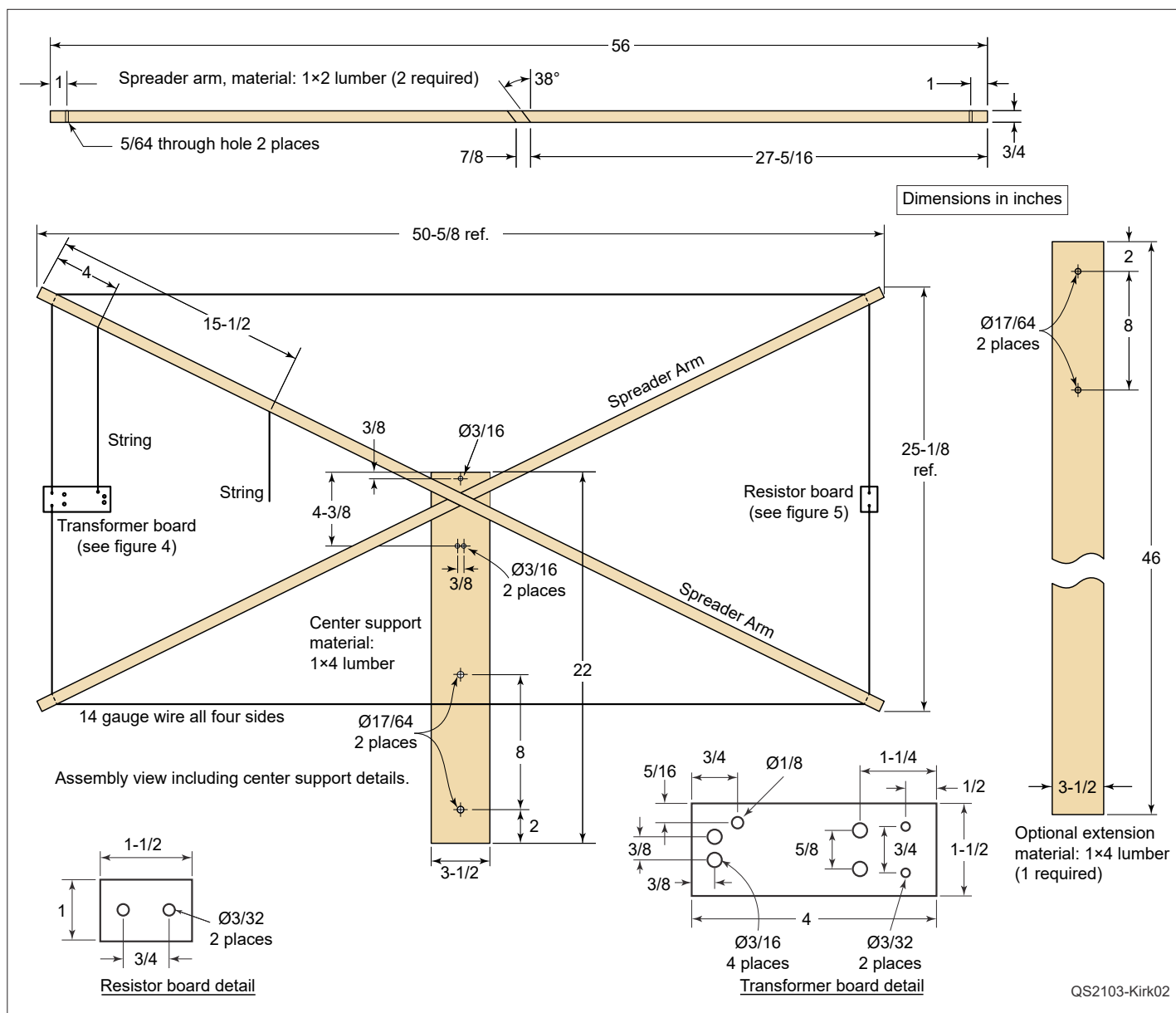


Figure 2 — The design sketch of the Portable Flag Antenna. Attach the extension to the center support using 1/4-20 nylon bolts.



Figure 3 — The clamp-on ferrite core with four passes of coax cable and shielded phono plug.

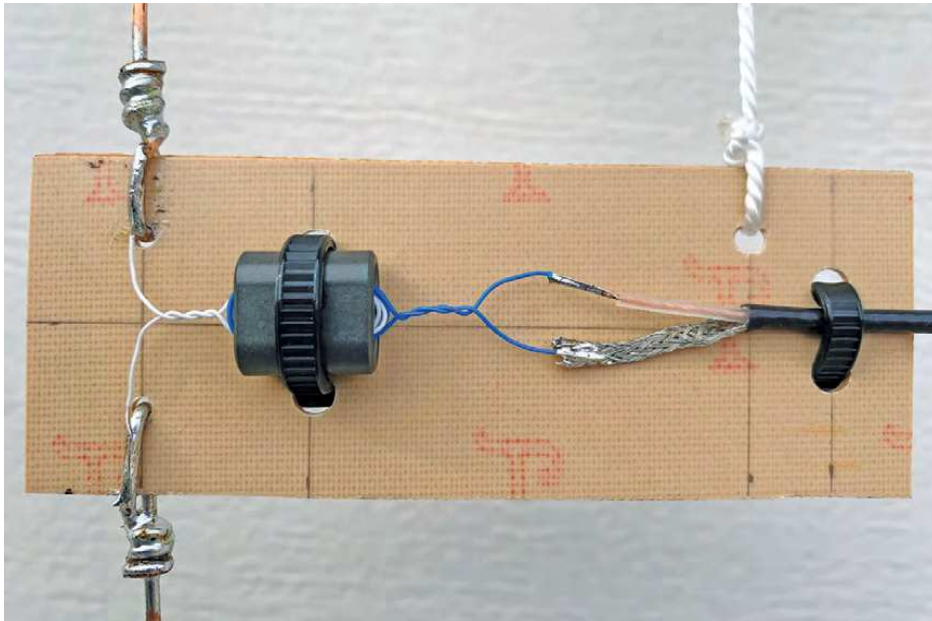


Figure 4 — The transformer board. Note the string support. Transformer three-turn primary connects to the coax.

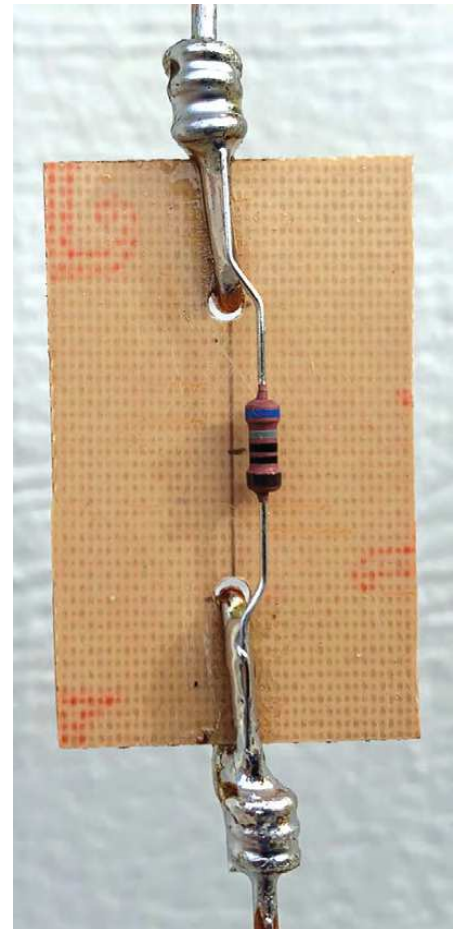


Figure 5 — The resistor is mounted to the board with some slack in the resistor legs to avoid mechanical stress on the resistor body.

I used a $\frac{1}{16}$ -inch-thick single-sided circuit board — with the copper removed — at the transformer end (see Figure 4) and at the resistor end (see Figure 5) of the antenna. One board is 1.5×4 inches and the other is 1×1.5 inches. I removed the copper using a rotary tool with a sanding band. Drill holes into the two boards, as seen in Figures 2, 4, and 5. Always wear safety glasses when using power tools.

The antenna comprises #14 AWG bare copper wire. When soldering the resistor to the antenna wire, leave some slack in the resistor legs to avoid mechanical stress on the resistor body. Mount the resistor board in the center of one of the 2-foot vertical wires; mount the transformer board in the center of the other 2-foot vertical wire. The resistor end of the antenna is the back of the antenna, while the transformer end is the front of the antenna.

Use zip ties to secure the transformer and coax to the transformer mounting board (see Figure 4), and to also secure the coax to the vertical center support board.

I used a piece of string to prevent sag in the transformer mounting board, and another piece of string to prevent sag in the horizontal run of coax. Drill a $\frac{1}{64}$ -inch-diameter hole 4 inches in from the spreader tip that's above the transformer board, and a second $\frac{1}{64}$ -inch-diameter hole 15.5 inches in from the same spreader tip for the coax support string. Drill both of these holes perpendicular to the spreader arm.

When soldering the resistor to the antenna wire, leave some slack in the resistor legs to avoid mechanical stress on the resistor body.

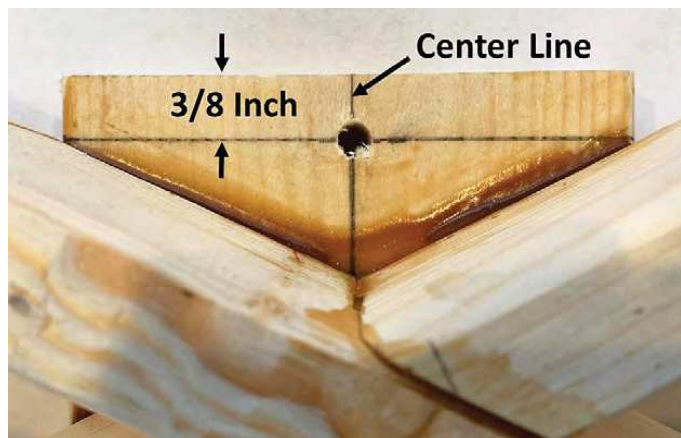


Figure 6 — The center support post showing alignment with the cross supports.

Frame Construction Details

Additional images of the lumber construction are available on the www.arrl.org/qst-in-depth web page. Cut two pieces of 1 × 2 lumber (actual size $\frac{3}{4} \times 1\frac{1}{2}$ inch) to a length of 56 inches to form the two spreader arms. I used a miter saw set at an angle of 38 degrees to cut a $\frac{7}{8}$ -inch-wide slot $27\frac{5}{16}$ inches in from the end of each board. I used a miter saw to remove the wood in the slots to a depth of at least halfway through each spreader arm. The slots allow the spreader arms to straddle each other when glued to the 1 × 4 (actual size $\frac{3}{4} \times 3\frac{1}{2}$ inch) center support board (see Figure 6).

Cut a piece of 1 × 4 lumber to a length of 22 inches for use as the center support, and another piece to a length of 46 inches for use as an optional extension seen next to my wife, Chris, in the lead photo.

Drill holes in each piece of wood, as shown in Figure 2. Drill the string support holes after everything is assembled.

Draw a line $\frac{3}{8}$ inches in from the top of the 22-inch center support board, and another line in the center of this board. Apply wood glue to the surfaces in the slot of each spreader arm. Also apply wood glue to the surfaces of each spreader arm that will come into contact with the center support (see Figure 6). Clamp the spreader arms to the center support, making sure the spreader arms line up with the middle of the center support line, as well as where the $\frac{3}{8}$ -inch line intersects the outer edges of the center support. The distance between the top of the upper spreader arm tip and the bottom of the lower spreader arm tip should be approximately 25 inches. I used Titebond III wood glue and allowed it to harden for 24 hours before removing the clamp.

Preamps

I use a W7IUV broadband 20 dB gain preamp, which may be available as a kit (www.mtmscientific.com/preamp.html), and it is also easy to homebrew (<https://sites.google.com/site/rxpreamps>).

On 160 and 80 meters (where the antenna gain is –63 and –52 dBi, respectively), I sometimes switch to a higher gain dual-band preamp to hear weak levels of RFI. That preamp, designed by Doug DeMaw, W1FB (explained in the August 1988 issue of *QST*), offers more than 40 dB of gain. A 9 V battery powers the W7IUV and W1FB preamps. They work just fine down to 7 V.

As you approach the source of RFI, you will likely need to reduce the system gain to detect a null. You accomplish this by removing the preamp, reducing receiver gain, or switching in attenuation.

Summary

The portable flag has become a real game changer, and it is my antenna of choice for direction finding of RFI on MF and HF. It is unidirectional, broadband, portable, easy to walk with while direction finding, and easy to transport in a car.

The portable flag does a wonderful job leading me to arcing power line poles, but when close to the suspect poles I switch to a portable 136 MHz AM receiver, four-element beam, and typically 30 dB of attenuation to identify the faulty pole.

DX Engineering (www.dxengineering.com) offers a complete kit part number DXE-NOISELOOP that is based on this design. They also offer a broadband 30 dB preamplifier and attenuator, part number DXE-NL-PRE-ATT-1, designed for use with the portable flag.

All photos by the author.

Don Kirk, WD8DSB, was first licensed in 1976 at age 16, and currently holds an Amateur Extra-class license as well as a General Radiotelephone Operator license. He received an associate degree in applied science from Henry Ford Community College in 1983 and a bachelor's degree in Engineering Technology (electrical/electronic) from Wayne State University in 1985. Don has been employed for the past 35 years as a senior engineer by Magnequench, working in the rare-earth permanent magnetic materials industry. Don's technical passion is discrete electronics and microcontroller-based projects, and he enjoys chasing DX and contesting on 160-meter CW. You can reach Don at wd8dsb@aol.com.

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